

Day 74 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in History & National Movement (4): 150 Years of Vande Mataram

READING MATERIAL

Covers the year-long 150th-anniversary commemoration of 'Vande Mataram' (2025-26) and the song's real history within the Indian National Movement.

Q: What nationwide commemoration began on 7 November 2025, and how long does it run?

A: A year-long commemoration of 150 years of 'Vande Mataram', running from 7 November 2025 to 7 November 2026 — approved by the Union Cabinet to connect citizens, especially youth and students, with the song's revolutionary spirit.

Q: What was the theme of Republic Day 2026, and which anniversary did it mark?

A: Republic Day 2026's theme was '150 Years of Vande Mataram' — marking the 77th anniversary of India's Constitution.

Q: Who composed 'Vande Mataram', when, and in which work did it first appear in print?

A: Bankim Chandra Chatterjee composed it in 1876 (in Bengali and Sanskrit); it was featured in his 1882 novel 'Anandamath', which is how it became widely known.

Q: When and by whom was 'Vande Mataram' first sung at an Indian National Congress session?

A: 28 September 1896, at the INC session in Calcutta, sung by Rabindranath Tagore.

Q: When did 'Vande Mataram' first become a political slogan, and in connection with which movement?

A: 7 August 1905, during student processions at Calcutta Town Hall — becoming the rallying cry of the Swadeshi movement and the protest against the Partition of Bengal.

Q: What is 'Vande Mataram's official constitutional status in India today?

A: It is India's National SONG (not the National Anthem — that is 'Jana Gana Mana') — given equal status with the National Anthem when India became a Republic in 1950.

MOCK TEST (WITH ANSWERS)

1. The 150th-anniversary commemoration of 'Vande Mataram' runs from 7 November 2025 to:

A. 7 November 2026 ✓

B. 26 January 2026

C. 15 August 2026

D. 7 August 2026

Explanation: 7 November 2026 (a year-long commemoration). [medium, web-research]

2. Republic Day 2026's theme was:

A. 75 Years of the Constitution

B. 150 Years of Vande Mataram ✓

C. Amrit Kaal

D. Digital India @10

Explanation: 150 Years of Vande Mataram. [medium, web-research]

3. 'Vande Mataram' was composed by Bankim Chandra Chatterjee in 1876, and later featured in his novel:

A. Durgeshnandini

B. Anandamath ✓

C. Devi Chaudhurani

D. Kapalkundala

Explanation: Anandamath (1882). [medium, web-research]

4. 'Vande Mataram' was first sung at an Indian National Congress session (Calcutta, 28 September 1896) by:

A. Bankim Chandra Chatterjee

B. Rabindranath Tagore ✓

C. Subhas Chandra Bose

D. Aurobindo Ghosh

Explanation: Rabindranath Tagore. [medium, web-research]

5. 'Vande Mataram' became a political slogan/rallying cry of which movement, starting 7 August 1905?

A. Non-Cooperation Movement

B. Quit India Movement

C. Swadeshi Movement (anti-Partition of Bengal) ✓

D. Civil Disobedience Movement

Explanation: Swadeshi Movement, protesting the Partition of Bengal. [medium, web-research]

6. 'Vande Mataram' holds what official status in India, distinct from the National Anthem?

A. It is the National Anthem itself

B. It is the National Song ✓

C. It has no official status

D. It is only a state song of West Bengal

Explanation: The National Song (Jana Gana Mana is the National Anthem). [easy, web-research]

Part B – Aptitude & Mental Ability: Time and Work

READING MATERIAL

Time and Work is the last of the 10 named Aptitude skills — after this, Part B moves to its Reasoning unit. Real exam evidence shows heavy use of 'trap' questions designed to look like they need complex math but actually resolve via a simple ratio.

Q: Method — inverse proportion (more workers, less time): 12 men complete a work in 36 days. In how many days will 18 men finish the same work?

A: Men $\times$ Days = constant (total work). $12 \times 36 = 18 \times x \rightarrow x = 432/18 = 24$ days. More workers means proportionally fewer days.

Q: Method — sequential combined work: A, B, C can complete a job alone in 8, 12, 16 days respectively. A and C work together for 2 days, then C leaves and B joins A. In how many more days do A and B finish?

A: Work rates: $A=1/8$, $B=1/12$, $C=1/16$ per day. In 2 days, A+C complete $2 \times (1/8 + 1/16) = 2 \times (3/16) = 3/8$ of the work. Remaining = $5/8$.
A+B's combined rate = $1/8 + 1/12 = 5/24$ per day. Days needed = $(5/8) \div (5/24) = (5/8) \times (24/5) = 3$ days.

Q: Method — wages split by share of work done: A can do a job in 10 days, B in 15 days. They work together for 5 days, then C finishes the remaining work in 2 days. Total payment for the whole job is Rs.5,000 — how should it be split?

A: Pay is split in proportion to how much of the TOTAL work each person actually did, not how many days they worked. In 5 days, A completes $5/10=1/2$ of the job, B completes $5/15=1/3$; together that's $5/6$, leaving $1/6$ for C to finish. Split of Rs.5,000 by work share: A gets $1/2 \times 5000 = \text{Rs.}2,500$; B gets $1/3 \times 5000 \approx \text{Rs.}1,667$; C gets $1/6 \times 5000 \approx \text{Rs.}833$. The key insight is the split follows WORK DONE, not days worked — C worked fewer days than B but did a comparable or different share depending on the rates, and pay tracks output, not time.

Q: Method — pairwise combined rates $\rightarrow$ solve for all three together: A+B finish a job in 12 days, B+C in 15 days, C+A in 20 days. In how many days can A, B, C finish it together?

A: Add all three pairwise rates: $(A+B) + (B+C) + (C+A) = 1/12 + 1/15 + 1/20 = 2(A+B+C)$'s rate. LCM=60: $5/60 + 4/60 + 3/60 = 12/60 = 1/5$.
So $2 \times (A+B+C \text{ rate}) = 1/5 \rightarrow (A+B+C) \text{ rate} = 1/10 \rightarrow$ together they finish in 10 days.

Q: Trap question: If 9 spiders make 9 webs in 9 days, how many days does 1 spider take to make 1 web?

A: 9 days — NOT 1 day. Each spider makes 1 web in 9 days regardless of how many spiders work in parallel (they're not collaborating on the SAME web). This tests whether you recognize the rate-per-individual stays fixed, the classic trap being to assume the answer scales down to 1.

Q: Method — pipes filling and draining together: Two pipes fill a tank in 30 and 40 minutes separately; a third pipe drains it in 24 minutes. With all three open, how long to fill the tank?

A: Combined rate = $1/30 + 1/40 - 1/24$ (subtract the draining pipe's rate). LCM=120: $4/120 + 3/120 - 5/120 = 2/120 = 1/60$. So the tank fills in 60 minutes.

Q: Method — provisions lasting fewer days with more people (inverse proportion again): 1,000 soldiers have 70 days of provisions. If 400 more soldiers join, how long will provisions last?

A: People $\times$ Days = constant (total provisions). $1000 \times 70 = 1400 \times x \rightarrow x = 70000/1400 = 50$ days.

Q: Method — find the second worker's solo time from a combined time and the first worker's solo time: A and B together finish a job in 16 days; A alone takes 48 days. How long does B alone take?

A: B's rate = $(A+B)$'s rate - A's rate = $1/16 - 1/48 = 3/48 - 1/48 = 2/48 = 1/24$. So B alone takes 24 days.

Q: Trap question: If 5 persons complete 5 projects in 5 days, how long do 50 persons take to complete 50 projects?

A: Still 5 days — NOT 50 days. The ratio of people to projects stays the same (1:1), so the per-project time doesn't change; 50 people working on 50 projects is just 10 independent copies of the original 5-person/5-project scenario happening in parallel.

MOCK TEST (WITH ANSWERS)

1. 15 men complete a work in 24 days. In how many days will 20 men finish the same work?

- A. 16 days
- B. 18 days ✓**
- C. 20 days
- D. 30 days

Explanation: $15 \times 24 = 20 \times x \rightarrow x = 360/20 = 18$ days. [easy, authored]

2. A and B together can finish a job in 18 days; A alone takes 45 days. How long does B alone take?

- A. 24 days

- B. 27 days
- C. 30 days ✓**
- D. 36 days

Explanation: B's rate = $\frac{1}{18} - \frac{1}{45} = \frac{5}{90} - \frac{2}{90} = \frac{3}{90} = \frac{1}{30}$. B alone takes 30 days. [medium, authored]

3. If 7 workers build 7 walls in 7 days, how many days does 1 worker take to build 1 wall?

- A. 1 day
- B. 7 days ✓**
- C. 14 days
- D. 49 days

Explanation: 7 days — each worker's individual rate is unaffected by how many work in parallel on separate walls. [medium, authored]

4. Two pipes fill a tank in 12 and 15 minutes separately; a third pipe drains it in 20 minutes. With all three open, how long to fill the tank?

- A. 8 minutes
- B. 10 minutes ✓**
- C. 12 minutes
- D. 15 minutes

Explanation: Rate = $\frac{1}{12} + \frac{1}{15} - \frac{1}{20}$. LCM=60: $\frac{5}{60} + \frac{4}{60} - \frac{3}{60} = \frac{6}{60} = \frac{1}{10}$. Time = 10 minutes. [medium, authored]

5. 800 soldiers have provisions for 60 days. If 200 more soldiers join, how long will the provisions last?

- A. 40 days
- B. 45 days
- C. 48 days ✓**
- D. 50 days

Explanation: $800 \times 60 = 1000 \times x \rightarrow x = 48000 / 1000 = 48$ days. [easy, authored]

6. A, B, and C can do a job alone in 24, 30, and 40 days respectively. Working together, how long will they take?

- A. 8 days
- B. 10 days ✓**
- C. 12 days
- D. 15 days

Explanation: Combined rate = $\frac{1}{24} + \frac{1}{30} + \frac{1}{40}$. LCM=120: $\frac{5}{120} + \frac{4}{120} + \frac{3}{120} = \frac{12}{120} = \frac{1}{10}$. Together they take 10 days. [medium, authored]

7. If 8 people complete 8 tasks in 8 days, how long do 40 people take to complete 40 tasks?

- A. 8 days ✓**
- B. 20 days
- C. 40 days
- D. 320 days

Explanation: Still 8 days — the people-to-task ratio (1:1) is unchanged, so per-task time is unchanged. [medium, authored]

Part C – General English: Grammar: Tenses (Simple, Continuous, Perfect, and Conditional Sentences)

READING MATERIAL

Real TNPSOC tense questions mostly test future-tense variants (matched to time clues) and conditional-sentence tense agreement — not abstract tense theory.

Q: Real exam: "The Doctor ____ the operation tomorrow." Which tense fits?

A: Will perform — simple future, matching the explicit future time clue 'tomorrow'.

Q: Real exam: "In two years' time I ____ (earn) my degree." Which tense fits?

A: Shall have earned — future perfect tense, used because the action (earning the degree) will be COMPLETED by a specific future point in time ('in two years' time').

Q: Real exam: "Will you be doing laundry tonight?" What tense form is this question itself in?

A: Future continuous tense ('will be doing') — describes an action expected to be in progress at a specific future time.

Q: Real exam: "She ____ an exam at 2 p.m. tomorrow." Which tense fits, and why not simple future?

A: Will be taking — future continuous, because the exam is an ongoing action happening AT a specific future time (2 p.m. tomorrow), not simply a one-off future event ('will take' would suggest a single, instantaneous action rather than an exam in progress at that time).

Q: Real exam: "The plane ____ by the time I got to the airport." Which tense fits?

A: Had left — past perfect tense, since the plane's departure happened BEFORE another past action (arriving at the airport) — the classic 'earlier of two past events' use of past perfect.

Q: Real exam (first conditional): "If she works hard, she ____ a scholarship." Which tense fits in the main clause?

A: Will get — first conditional structure: 'If' + simple present (works) → simple future (will get), used for a realistic/likely future outcome.

Q: Real exam (third conditional): "If you ____ hard you would have won the relay match." Which tense fits in the if-clause?

A: Had worked — third conditional structure: 'If' + past perfect (had worked) → 'would have' + past participle (would have won), used for an unreal/hypothetical PAST situation (the relay match is already over and was NOT won).

MOCK TEST (WITH ANSWERS)

1. "The Doctor ____ the operation tomorrow."

A. Perform

B. Will perform ✓

C. Performed

D. Performing

Explanation: Will perform. [easy, question-bank-2022]

2. "In two years' time I ____ (earn) my degree."

A. Have earned

B. Shall have earned ✓

C. Shall earn

D. Have been earning

Explanation: Shall have earned (future perfect). [medium, question-bank-2022]

3. "Will you be doing laundry tonight?" is in which tense?

A. Future perfect tense

B. Future tense

C. Future continuous tense ✓

D. Future perfect continuous tense

Explanation: Future continuous tense. [medium, question-bank-2025]

4. "She ____ an exam at 2 p.m. tomorrow."

A. will take

B. will be taking ✓

- C. shall take
- D. shall be taking

Explanation: will be taking (future continuous). [medium, question-bank-2025]

5. "The plane ____ by the time I got to the airport."

- A. is left
- B. had left ✓**
- C. was left
- D. were left

Explanation: had left (past perfect). [medium, question-bank-2025]

6. "If she works hard, she ____ a scholarship."

- A. would get
- B. would have got
- C. will get ✓**
- D. won't get

Explanation: will get (first conditional). [medium, question-bank-2024]

7. "If you ____ hard you would have won the relay match."

- A. will work
- B. works
- C. working
- D. had worked ✓**

Explanation: had worked (third conditional). [hard, question-bank-2024]